

# Future Mall Smart Garage - Circuit and Code Reference

Portfolio ID: EYOUTH-30909090117044

This document shows the exact ESP32 wiring used in the included Wokwi project and the MicroPython program source. It is included as a circuit/code reference; a Wokwi simulation screenshot and share link must be made after importing the project into Wokwi.

| Part         | GPIO pin |
|--------------|----------|
| Entry button | GPIO 12  |
| Exit button  | GPIO 13  |
| Green LED    | GPIO 25  |
| Red LED      | GPIO 26  |
| Yellow LED   | GPIO 27  |
| Blue LED     | GPIO 14  |

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# Future Mall Smart Garage for ESP32 MicroPython
# Project ID: EYOUTH-30909090117044
# Entry and exit buttons use INPUT_PULLUP, so a pressed button reads 0.

from machine import Pin
from time import sleep

entry_button = Pin(12, Pin.IN, Pin.PULL_UP)
exit_button = Pin(13, Pin.IN, Pin.PULL_UP)

green_led = Pin(25, Pin.OUT) # Spaces available
red_led = Pin(26, Pin.OUT)   # Garage full
yellow_led = Pin(27, Pin.OUT) # Car entered
blue_led = Pin(14, Pin.OUT)  # Car exited

car_count = 0
MAX_CARS = 15

def show_garage_status():
    """Show whether the garage has a free space."""
    if car_count < MAX_CARS:
        green_led.on()
        red_led.off()
    else:
        green_led.off()
        red_led.on()

def blink(led):
    """Blink one LED once to show a car movement."""
    led.on()
    sleep(0.3)
    led.off()

show_garage_status()

while True:
    if entry_button.value() == 0:
        # Do not let the count become more than 15.
        if car_count < MAX_CARS:
            car_count = car_count + 1
            print("Car entered. Cars in garage:", car_count)
            blink(yellow_led)
        else:
            print("Garage is full")

        show_garage_status()
        # Wait for the button to be released, so one press adds one car.
        while entry_button.value() == 0:
            sleep(0.1)

    if exit_button.value() == 0:
        # Do not let the count become less than 0.
        if car_count > 0:
            car_count = car_count - 1
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        print("Car exited. Cars in garage:", car_count)
        blink(blue_led)
    else:
        print("Garage is empty")

    show_garage_status()
    # Wait for the button to be released, so one press removes one car.
    while exit_button.value() == 0:
        sleep(0.1)

sleep(0.1)
```